C MONTE CARLO RELATIVE IMPROVEMENTS

The algorithm described in Section 2.3 can be formalized as:

Algorithm 1: Expected worst-case relative improvement

Input: Step size α
Result: Mean of relative length improvements

$L \leftarrow$ empty list;
while *maximal samples not reached* **do**
  $\boldsymbol{x}_A \leftarrow$ sample latent vector;
  $\boldsymbol{v}_{max} \leftarrow$ eigenvector of metric with largest eigenvalue at $\boldsymbol{x}_A$;
  $\boldsymbol{x}_B \leftarrow \boldsymbol{x}_A + \alpha \boldsymbol{v}_{max}$;

  $d_s \leftarrow$ compute Riemannian length of straight line;
  $d_c \leftarrow$ find shorter length;
  $relative_improvement \leftarrow \frac{d_s - d_c}{d_s}$;
  Append $relative_improvement$ to L;
end

return *mean of* L;

For a concrete example, we ran it on a trained VAE for MNIST digits. After acquiring a list of relative improvements, we plot the frequency of occurrence in certain intervals, to get a good idea of how much the relative improvement varies in magnitude, see Figure 4.

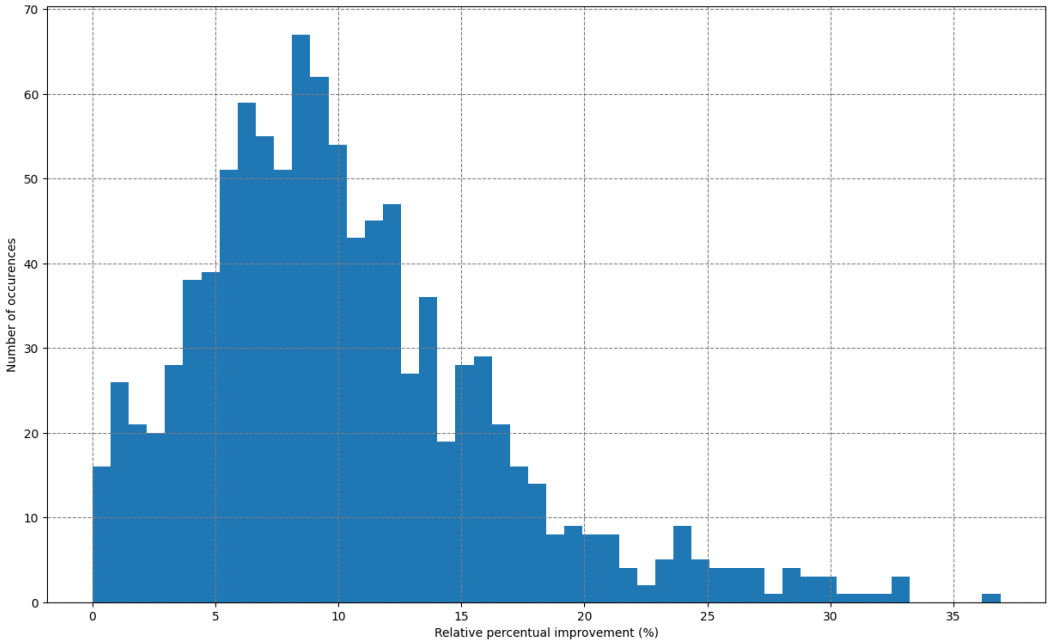

Figure 4: Histogram of the distribution of relative length improvements for a VAE latent space using 1000 samples according to Algorithm 1.

D COMPARABILITY OF LATENT LINEAR INTERPOLATIONS: EXAMPLE STRATEGY

In the following we trained two VAEs with different architecture (both 2D latent space) on MNIST digits 2,4,5 and 7, and compared their interpolation capability as described in Section 3. VAE1 has 16 times more trainable parameters than VAE2. We random sampled 20 pairs of start and end points in input space, which can be seen in Figure 5. We then found the relative improvements that are possible for each pair using both models, which can be seen in Figure 6.

Figure 5: Start and end points of 20 input space samples.

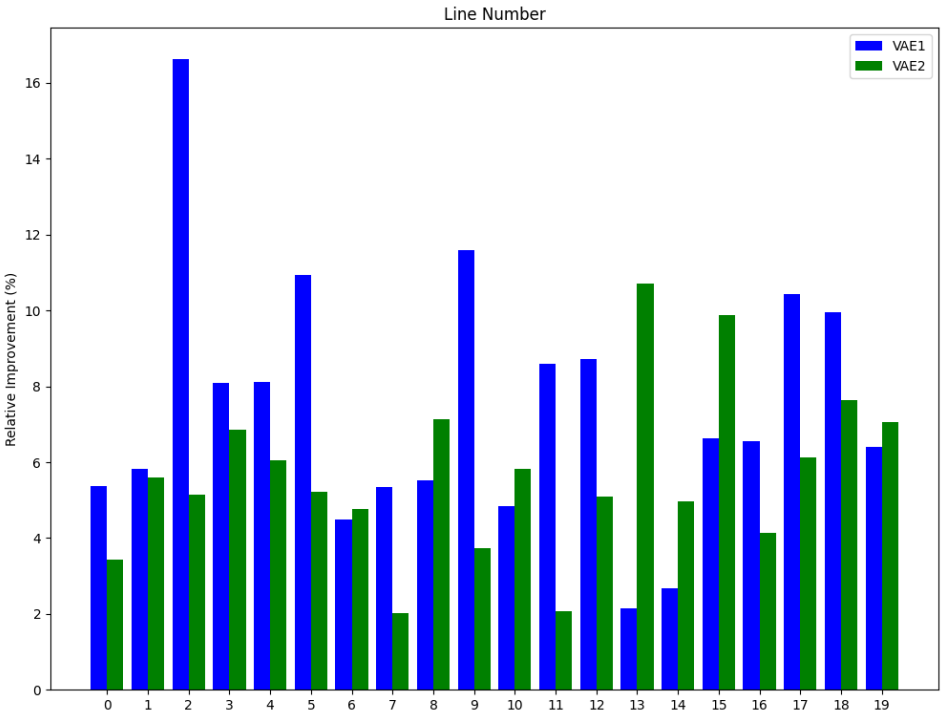

Figure 6: Bar plot of relative improvements for 20 random sampled latent linear interpolation for two VAEs.

We observe that the relative improvements of both models can be vastly different, mainly due to their latent spaces being shaped differently (see Figure 7). Therefore interpolations in similar metric neighborhoods should be compared, and from the above samples we could, for instance, choose lines 1, 6, 19 to represent a fair comparison (a fixed threshold should be chosen). One of those interpolations can be seen in Figure 8.

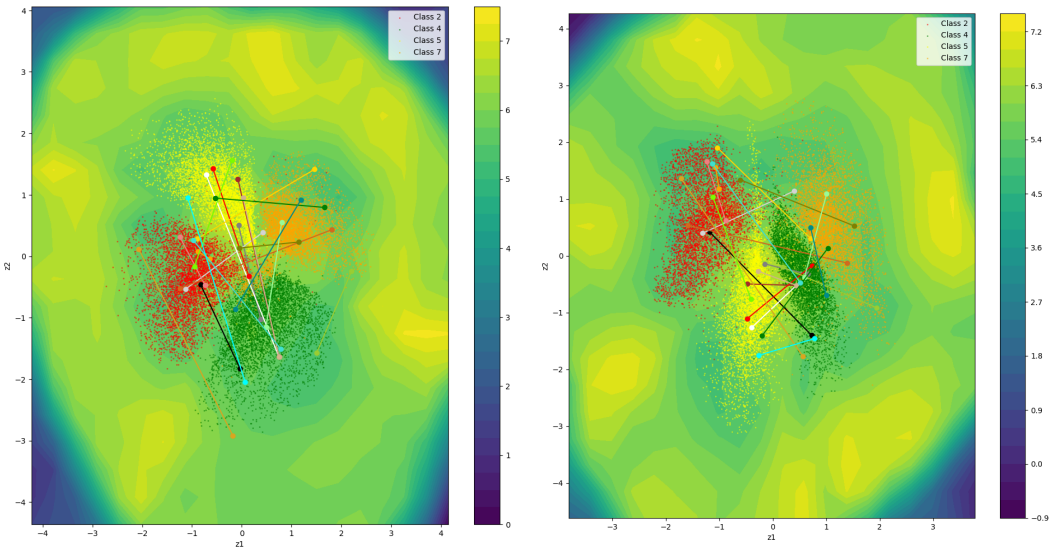

Figure 7: VAE1 can be seen on the left, VAE2 on the right. Both use the improved variance estimate of Arvanitidis et al. (2018). The colored lines are the 20 interpolation samples, same colors are used in both figures.

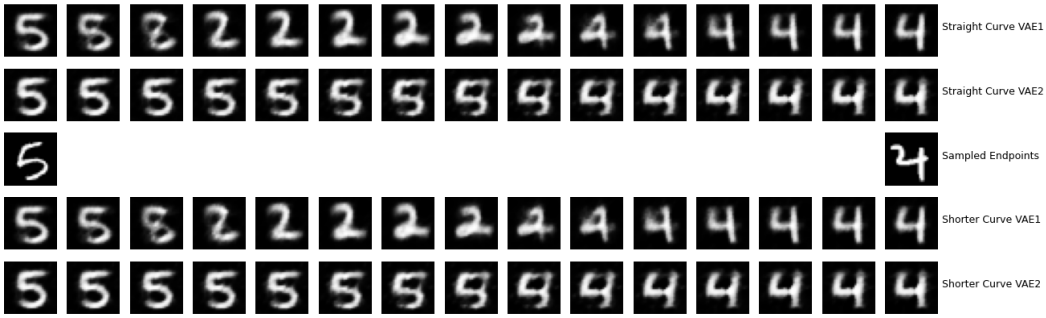

Figure 8: Interpolation sequence 19. The original images for start and end point are displayed in the middle row.

E Feature Mapping to new Output Space: Induced Metric

Next we derive the effect that chaining a function on a VAE has on the resulting Jacobian/induced metric. We define the decoder of our VAE as:

$$\phi(\boldsymbol{z}) = \boldsymbol{\mu}(\boldsymbol{z}) + \boldsymbol{\sigma}(\boldsymbol{z}) \cdot \boldsymbol{\epsilon}$$

Where we have $\boldsymbol{z} \in \mathbb{R}^d$, $\phi : \mathbb{R}^d \mapsto \mathbb{R}^D$ and $\boldsymbol{\epsilon} \in \mathcal{N}(\mathbf{0}, \mathbb{I}_D)$ (zero-centered multivariate normal distribution with identity matrix covariance).

The result of the decoder of the VAE (with same dimension as input space X) then is mapped through some function $h : \mathbb{R}^D \mapsto \mathbb{R}^K$ into the new output space. In total we apply the function f on latent input $\boldsymbol{z}$, with Jacobian:

$$f(\boldsymbol{z}) = h(\boldsymbol{\mu}(\boldsymbol{z}) + \boldsymbol{\sigma}(\boldsymbol{z}) \cdot \boldsymbol{\epsilon})$$

$$\mathbf{J}_{Fz} := \frac{\partial f(\boldsymbol{z})}{\partial \boldsymbol{z}} = \frac{\partial h(\phi)}{\partial \phi} \frac{\partial \phi(\boldsymbol{z})}{\partial \boldsymbol{z}} =: \mathbf{J}_{Fx} \mathbf{J}_{xz}$$

Dimensions: $\mathbf{J}_{Fz} \in \mathbb{R}^{K \times d}$, $\mathbf{J}_{Fx} \in \mathbb{R}^{K \times D}$, $\mathbf{J}_{xz} \in \mathbb{R}^{D \times d}$. Note: the functions until now should be defined in such a way that the Jacobians are nonsingular.